

GOURE, Niger

WMO: 610450

Lat: 13.983N

Lon: 10.300E

Elev: 1509

StdP: 13.91

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	54.7	56.9	17.2	13.9	78.7	17.4	14.0	79.5	20.3	71.1	18.4	69.3	2.3	40	0.375

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	24.0	107.9	67.9	106.4	67.7	104.8	67.7	78.2	93.1	77.3	91.6	76.6	90.5	4.8	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.6	137.2	83.4	73.8	133.3	82.9	73.1	130.1	82.8	42.8	93.0	41.9	91.7	41.3	90.3	86.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.7	13.9	11.6	DB	50.0	111.0	1.8	1.1	48.7	111.8	47.6	112.5	46.6	113.1	45.3	113.9	
			WB	38.5	81.4	1.2	3.2	37.7	83.7	37.0	85.6	36.3	87.4	35.4	89.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	83.6	70.8	76.7	84.5	91.0	93.2	91.5	86.5	83.3	86.1	86.8	79.6	72.3
	DBStd	8.11	5.08	5.90	4.69	3.66	3.12	3.94	3.71	3.40	3.39	2.89	4.06	4.20
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	5	5	0	0	0	0	0	0	0	0	0	0	0
	CDD50	12249	646	748	1069	1230	1340	1245	1133	1034	1084	1139	889	692
	CDD65	6780	186	328	604	780	875	795	668	569	634	674	439	228
	CDH74	91968	1932	3750	8258	12119	14063	12060	8817	6512	8005	9141	5086	2225
	CDH80	54611	768	1917	5005	8161	9721	7881	4828	3023	4322	5414	2717	854
Wind	WSAvg	4.1	4.6	4.2	4.0	3.8	4.5	5.7	5.4	3.5	2.5	2.9	3.5	4.1
Precipitation	PrecAvg	11.1	0.0	0.0	0.0	0.0	0.3	1.0	3.7	4.7	1.9	0.2	0.0	0.0
	PrecMax	19.6	0.0	0.0	0.1	1.0	1.4	4.0	9.3	11.5	6.1	1.6	0.0	0.0
	PrecMin	3.5	0.0	0.0	0.0	0.0	0.0	0.0	0.7	0.2	0.1	0.0	0.0	0.0
	PrecStd	3.7	0.0	0.0	0.0	0.2	0.4	0.9	1.9	2.3	1.5	0.4	0.0	0.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	95.9	101.1	105.4	109.3	110.1	108.0	103.4	98.2	103.0	103.8	99.9	95.4
		MCWB	60.5	63.3	63.3	67.4	68.3	70.5	71.6	76.3	68.8	67.3	65.0	61.3
	2%	DB	91.7	98.0	103.3	107.6	108.3	106.1	100.3	95.8	100.9	102.1	97.9	91.2
		MCWB	58.5	61.2	63.5	66.9	68.0	70.6	73.3	76.3	70.4	67.0	63.8	59.9
	5%	DB	88.3	94.7	101.1	106.0	106.9	104.3	97.7	93.8	98.5	100.4	95.5	88.1
		MCWB	57.2	60.0	62.6	66.2	67.9	71.2	74.0	76.1	72.3	67.2	62.3	58.4
	10%	DB	84.5	91.0	98.4	104.1	105.0	102.0	95.4	91.8	96.1	98.6	92.9	85.0
		MCWB	55.6	58.2	61.4	65.8	67.9	72.0	74.3	75.7	73.4	66.5	60.9	56.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	62.9	67.3	70.2	79.2	78.2	79.1	78.9	79.0	78.6	77.6	68.9	63.8
		MCDB	91.2	92.6	100.2	103.8	101.6	97.0	93.2	91.5	92.4	91.7	94.6	88.9
	2%	WB	59.7	62.8	67.0	74.2	75.5	76.8	77.2	77.8	77.7	75.6	65.6	61.5
		MCDB	89.3	94.5	96.3	98.6	97.3	95.2	91.3	90.4	91.3	90.2	94.6	87.7
	5%	WB	57.7	60.7	64.3	71.2	74.4	75.7	76.4	77.1	76.9	74.0	63.6	59.5
		MCDB	86.3	92.0	96.4	96.1	95.8	94.2	90.3	89.7	90.3	88.8	93.1	85.1
	10%	WB	55.8	58.8	62.4	68.8	73.3	74.8	75.6	76.5	76.2	72.5	61.8	57.5
		MCDB	83.3	90.0	95.9	96.2	94.7	93.3	89.4	88.3	89.0	87.6	90.8	83.2
Mean Daily Temperature Range	5% DB	MDBR	26.0	26.5	26.5	26.1	24.0	20.5	18.0	16.5	20.0	23.9	26.2	25.6
		MCDBR	30.4	29.9	29.5	27.6	26.3	22.8	20.4	18.8	23.0	25.8	28.2	28.7
		MCWBR	12.9	11.7	11.4	10.7	9.6	8.1	5.6	5.4	7.4	8.7	10.9	12.0
	5% WB	MCDBR	29.6	29.1	27.2	22.9	21.6	19.9	17.9	17.5	18.8	21.3	27.2	27.7
		MCWBR	13.2	12.2	11.9	11.4	8.4	7.0	5.9	5.9	6.7	8.1	11.7	12.8
Clear-Sky Solar Irradiance	taub		0.570	0.626	0.679	0.721	0.708	0.705	0.663	0.622	0.629	0.660	0.567	0.559
	taud		1.673	1.548	1.462	1.423	1.461	1.521	1.652	1.736	1.698	1.567	1.660	1.687
	Ebn at Noon		231	225	218	209	208	206	216	227	226	214	229	229
	Edh at Noon		76	89	100	103	98	91	80	75	78	87	76	73
All-Sky Solar Radiation	RadAvg		1793	1940	2073	2127	2131	2033	1937	1906	2003	1940	1834	1732
	RadStd		54	43	92	72	82	83	52	67	44	61	52	56

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page